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Geometrical optimization of packed-bed and parallel-plate active magnetic regenerators

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ABSTRACT

The paper presents the numerical optimization of a packed-bed and a parallel-plate *AMR* with gadolinium as the magnetocaloric material. The diameter of the spheres and the length of the packed-bed *AMR*, and the plate thickness, the spacing between the plates (porosity) and the length of the parallel-plate *AMR*, were varied and optimized. The results reveal that for the considered conditions the optimum length of the packed-bed *AMR* should be 20 mm or less at higher operating frequency (3 Hz) and 40 mm at lower operating frequency (0.5 Hz) and the corresponding optimum sphere diameter should be between 0.07 mm (with regard to the cooling load) and 0.17 mm (with regard to the *COP*). The parallel-plate *AMR* should consist of plates that are as thin as possible, with a spacing between the plates of 0.035 mm (with regard to the cooling load) and 0.075 mm (with regard to the *COP*).

Optimisation géométrique des régénérateurs actifs magnétiques à plaques parallèles et à lit fixe tassé

Mots clés : réfrigérateur magnétique ; régénérateur ; gadolinium ; modélisation ; optimisation

1. Introduction

A heat regenerator is a type of heat exchanger in the form of a porous structure through which two heat-transfer fluids are pumped periodically, usually in a counter-flow direction. The heat-transfer process is performed by the heat accumulation of a regenerative material (matrix). Heat regenerators have been widely used and analyzed for almost a century. As summarized and described in, e.g., Razelos (1979), the first mathematical model of a heat regenerator was developed in 1926 by Anzelious and a few years later by Nusselt, Schumann and Hausen. In later years, the increasing need for heat regenerators in different industrial and domestic applications led to an increased interest in the analysis and optimization of such heat exchangers. The mathematical model of a heat regenerator and its analytical and numerical solutions can be found in the literature (Coppage and London, 1953; Hill and Willmott, 1987; Foumney and Heggs, 1991; Dragutinovic and Baclic, 1998; Shah and Sekulic, 2003).

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Nomenclature

Variables

a	height of the *AMR* [m]
A_{ht}	total heat-transfer area [m^2]
c	specific heat [J kg^{-1} K^{-1}]
B	internal magnetic field [T]
COP	coefficient of performance [–]
d_h	hydraulic diameter [m]
d	sphere diameter/plate thickness [m]
f	friction-factor [–]
Gz	Graetz number [–]
H	internal magnetic field intensity [A m^{-1}]
L	length of the *AMR* [m]
$\dot{m}$	mass-flow rate [kg s^{-1}]
m	mass [kg]
Nu	Nusselt number [–]
P_f	duration of the fluid flow period (in one direction) [s]
P_{mag}	duration of the (de)magnetization period [s]
Pr	Prandtl number [–]
q	specific cooling load [W kg^{-1}]
Re	Reynolds number [–]
r	plate spacing [m]
$\check{s}$	width of the *AMR* [m]
T	temperature [K]
ΔT_{ad}	adiabatic temperature change [K]
ΔT	temperature span [K]
V_{AMR}	volume of the *AMR*
υ	fluid velocity [m s^{-1}]

Greek letters

α	heat-transfer coefficient [W m^{-2} K^{-1}]
ε	porosity [–]
λ	thermal conductivity [W m^{-1} K^{-1}]
μ_0	vacuum permeability [V s A^{-1} m^{-1}]
ν	frequency [Hz]
ρ	density [kg m^{-3}]
τ	residence time of the fluid in the *AMR* ($\tau = L/\upsilon$) [s]

Indices

c	cold
eff	effective
f	fluid
h	hot
H	constant magnetic field
s	solid

The development of magnetic refrigeration at room temperature opens up a new and relatively unexplored area where heat regenerators have a crucial role. Magnetic refrigeration is an emerging technology with a great potential to become competitive with the nowadays widely used vapour-compression refrigeration technology. It is based on the exploitation of the magnetocaloric effect, which causes the magnetocaloric material (i.e., a magnetic material with a pronounced magnetocaloric effect) to heat up when inserted into the magnetic field and to cool down when removed from the field. However, since the magnetocaloric effect of currently known magnetocaloric materials at room temperature is limited to a few Kelvins (for a magnetic field achievable with permanent magnets) a heat-regeneration process is required in order to enlarge the temperature span (the difference between the heat source and heat sink temperature) of the potential magnetocaloric device. Among the different types of heat regenerators, in recent years the active magnetic regenerator (*AMR*) has shown the greatest potential for applications in a magnetic refrigerator working at room temperature. The *AMR* is a heat regenerator that uses a magnetocaloric material as a regenerative material. It has a double function in the magnetic refrigerator: it works as a refrigerant, since it contains a magnetocaloric material; and it works as a heat regenerator and makes it possible to establish a temperature gradient along its length.

By 2010 as many as 41 prototypes of magnetic refrigerators had been designed and built (Yu et al., 2010). In general, all of these prototypes are based on an *AMR*. As we also know from the basic operational principle of heat regenerators (Shah and Sekulic, 2003), the *AMR*s and consequently the prototypes of the magnetic refrigerator can be divided into rotating devices (rotary movement of the *AMR* over a static magnetic field or vice versa (e.g., Zimm et al., 2006; Okamura et al., 2006; Tušek et al., 2010; Engelbrecht et al., 2012)) and reciprocating devices with a fixed or reciprocating regenerator bed (linear movement of the *AMR* or of the magnetic field (e.g. Bahl et al., 2008; Tura and Rowe, 2011; Tušek et al., 2013)). Regardless of its operational principle, the thermodynamic cycle of the *AMR* consists of the four basic operational processes:

A. Magnetization – the *AMR* is subjected to an increase in the magnetic field, which due to the magnetocaloric effect causes heating of the magnetocaloric material in the *AMR*.
B. Fluid flow from the cold external heat exchanger (CHEX) through the heated magnetocaloric material in the *AMR* into the hot external heat exchanger (HHEX), where this heated fluid rejects the heat to the surroundings.
C. Demagnetization – the *AMR* is subjected to a decrease of the magnetic field, which due to the magnetocaloric effect causes cooling of the magnetocaloric material in the *AMR*.
D. Fluid flow from the HHEX through the magnetocaloric material in the *AMR* into the CHEX (in the counter-flow direction with respect to process B.), where this cooler fluid absorbs the heat from the surroundings.

In general, all the prototypes of magnetic refrigerators built up to date so far use an *AMR* in the form of a packed-bed structure (spheres or irregular particles) or parallel-plate structure (Yu et al., 2010). The geometries that would be better from the thermo-hydraulic point of view (e.g., corrugated plates, honeycomb structures) are difficult to manufacture with current magnetocaloric materials and technologies.

In recent years gadolinium (Gd) has been accepted by the scientific community as the reference magnetocaloric material for room temperature applications. Compared to other magnetocaloric materials it has relatively good manufacturing properties and its magnetocaloric properties can be theoretically estimated using the mean-field theory (Morrish, 1965). The great majority of magnetic refrigerator prototypes as well as theoretical (numerical) analyses of the *AMR* use gadolinium as the magnetocaloric material. However, in recent years other materials have also shown great potential for use in *AMR*s, e.g., alloys based on LaFeSi (Bjork et al., 2010; Morrison et al., 2012) and MnFeP (Bruck et al., 2008). The geometry of the *AMR* with these materials is currently limited to powders (Yu et al., 2010) and plates with a minimum thickness and spacing of 0.25 mm and 0.2 mm, respectively (Katter et al., 2010).

This paper presents a geometrical optimization (based on a previously developed and experimentally verified numerical model (Tušek et al., 2011, 2012)) of all the crucial geometrical properties of packed-bed (with spheres) and parallel-plate *AMR*s with gadolinium as the magnetocaloric material. Since the assumption of a homogenous magnetic field in the material is applied (demagnetization effects are neglected) the optimization is performed from the thermo-hydraulic point of view.

2. Mathematical and numerical model of the AMR

In order to understand the operation of the *AMR* and its properties in detail a number of time-dependent mathematical and numerical models have been developed (Nielsen et al., 2011). These models serve for the simulation of the dynamic behavior of the *AMR* and the estimation of its cooling characteristics. A great majority of the developed models are one-dimensional (1-D), e.g., Bouchekara et al., 2008; Tagliafico et al., 2010; Risser et al., 2010; Vuarnoz and Kawanami, 2012. Recently, two-dimensional (2-D) models of the *AMR*, which consider also the dimension perpendicular to the fluid flow, were developed by, e.g., Nielsen et al., 2009; Liu and Yu, 2011; Oliveira et al., 2012. However, due to a random particle distribution in the packed-bed of the *AMR*, 2-D models can only be applied for *AMR*s with ordered structures (parallel-plate, square channel, etc.). Compared to the mathematical model of a classic heat regenerator the *AMR*'s mathematical model must additionally consider the magnetocaloric effect and the related temperature and magnetic field dependence of the specific heat of a magnetocaloric material.

For the purpose of this analysis the one-dimensional numerical model of the *AMR* presented in Tušek et al. (2011) was used. Initially, the numerical model was based on the assumption of an "on–off" magnetic field change and discrete implementation of the magnetocaloric effect. However, for the purpose of this study the numerical model was additionally upgraded with a built-in magnetocaloric effect in the differential equation of the magnetocaloric material (the second term on the left-hand side in Eq. (1)). The governing equation of the *AMR*'s mathematical model can be written as follows:

$$\frac{\partial T_s}{\partial t} + \frac{\partial(dT_{ad}(T_s,\mu_0 H))}{\partial \mu_0 H}\cdot\frac{d\mu_0 H}{dt} = \frac{\lambda_{eff}}{\rho_s\cdot c_s(T_s,\mu_0 H)\cdot(1-\varepsilon)}\frac{\partial^2 T_s}{\partial x^2} + \frac{\alpha_{eff}\cdot A_{ht}}{c_s(T_s,\mu_0 H)\cdot m_s}\cdot(T_f - T_s) \tag{1}$$

$$\frac{\partial T_f}{\partial t} + \frac{L}{\tau}\cdot\frac{\partial T_f}{\partial x} = \left| f\cdot\frac{2\cdot\left(\frac{L}{\tau}\right)^3}{d_h\cdot c_f}\right| + \frac{\alpha_{eff}\cdot A_{ht}}{c_f\cdot \dot{m}_f\cdot\tau}\cdot(T_s - T_f) \tag{2}$$

A detailed explanation of the governing equations, the initial and boundary conditions, the assumptions considered in the mathematical model and the numerical solution of the model are presented in Tušek et al. (2011). As already explained, the only exception is the assumption of an "on–-off" magnetic field change, which was for the purpose of this analysis replaced with the assumption of a time-dependent magnetic profile (linear increasing and decreasing of the magnetic field). The most relevant assumptions for this work are the following:

- The heat losses or gains from/to the surroundings are neglected;
- The geometry of the magnetocaloric material is uniformly distributed and the fluid flow through the *AMR* is homogenous – no fluid flow maldistribution occurs (for the packed-bed and the parallel-plate regenerators, the effect of maldistribution is studied in detail, e.g., duToit (2008) and Nielsen et al. (2012b), respectively);
- An ideal valve system is assumed, so the change of the inlet mass-flow rate is a discrete "on–off" function;
- The external heat exchangers are considered to be ideal;
- The magnetic field in the magnetocaloric material is homogenously distributed and equal to the applied magnetic field – demagnetization effects are neglected (for details on the impact of the demagnetization effect on the performance of the *AMR* see Nielsen et al., 2012a).

In the model the thermal conduction of the solid (magnetocaloric material) and the fluid is combined into the effective thermal conduction (λ_{eff}) and included it in the equation for the solid, like it is presented in, e.g., Kaviany (1995). Such an approach simplifies the equation for the fluid and also the numerical solution. Furthermore, because the 1-D model neglects the temperature distribution in the solid perpendicular to the fluid flow the correction factor for the heat-transfer coefficient needs to be applied (the effective heat-transfer coefficient (α_{eff})). Using such an approach the effect of the non-uniform temperature distribution is, to a certain extent, taken into account in the 1-D model as well (Engelbrecht et al., 2006).

The partial derivative of the adiabatic temperature change with respect to the magnetic field (the second term on the left-hand side in Eq. (1)) and the specific heat (c_H) of gadolinium included in the mathematical model are shown in Fig. 1. Both these properties were estimated by the mean-field theory (Morrish, 1965). The magnetic properties of gadolinium used as the input data of the mean-field theory were obtained from Tishin and Spichkin (2003).

The geometry of the *AMR* in the 1-D numerical models is considered through the correlations of the heat-transfer (α) and the friction-factor (f) coefficients. For the determination of the heat-transfer coefficient in the packed-bed *AMR* the Nusselt correlation presented by Wakao and Kaguei (1982) was used (Eq. (3)), while for the parallel-plate *AMR* the Nusselt correlation presented by Nickolay and Martin (2002) was used (Eq. (4)).

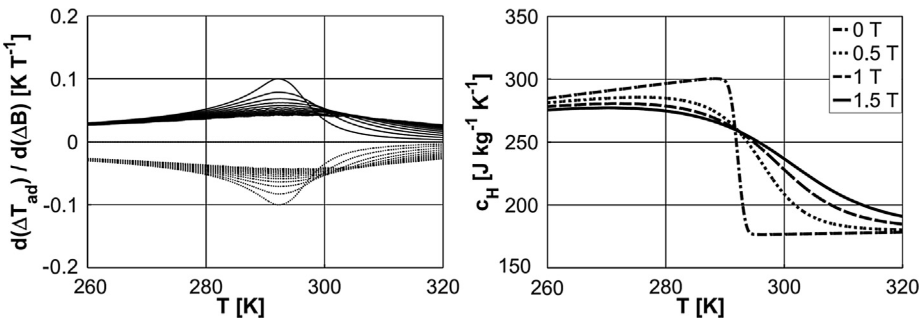

Fig. 1 – Left: The partial derivative of the adiabatic temperature change of gadolinium with respect to the magnetic field for the magnetization process (full lines) and the demagnetization process (dotted lines) in the case of a step of the magnetic field change of 0.2 T and a final magnetic field change of 2 T; Right: Specific heat of gadolinium at different magnetic fields.

$$Nu = 2 + 1.1 \cdot \mathrm{Re}^{0.6} \cdot \mathrm{Pr}^{\frac{1}{3}} \quad (10 < \mathrm{Re} < 10^4), \tag{3}$$

$$Nu = \left(7.541^{3.592} + \left(1.841 \cdot Gz^{\frac{1}{3}}\right)^{3.592}\right)^{\frac{1}{3.592}} \quad (Gz < 10^5). \tag{4}$$

For the friction-factor correlations the correlations presented by Kays and London (1984) were used for the packed-bed (Eq. (5)) and for parallel-plate (Eq. (6)) *AMR*.

$$f = 23.462 \cdot \mathrm{Re}^{-0.6716} \quad (10 < \mathrm{Re} < 5 \cdot 10^5), \tag{5}$$

$$f = 24/\mathrm{Re} \quad (\mathrm{Re} < 2300). \tag{6}$$

The hydraulic diameter of both types of analyzed *AMR*s was calculated based on the following equation:

$$d_h = \frac{4 \cdot V_{AMR} \cdot \varepsilon}{A_{ht}}. \tag{7}$$

Above-presented correlations of the Nusselt number and the friction-factor have been commonly used in 1-D numerical models of the *AMR* (see e.g., Petersen et al., 2008; Engelbrecht et al., 2005; Tagliafico et al., 2010).

3. Results and discussion

In addition to the *AMR*'s geometry, its operation is also strongly affected by its operational parameters. The operating frequency (the number of thermodynamic cycles per unit of time) and the mass-flow rate through the *AMR* have a very important impact on the generated cooling load and the *COP* of the refrigeration cycle, as recently shown by, e.g., Tagliafico et al. (2010), Tušek et al. (2011) and Vuarnoz and Kawanami (2012). If the operational parameters are not chosen appropriately (also regarding the *AMR*'s other properties, like geometry, magnetic field change, magnetocaloric material, heat-transfer fluid, etc.) the *AMR*'s cooling characteristics would be significantly reduced. This is the reason for optimizing the *AMR*'s geometrical properties at different mass-flow rates and for different operating frequencies.

3.1. *Analyzed operational and geometrical parameters*

The optimization of the *AMR*'s geometry was performed for two typical *AMR*s (packed-bed of spheres and parallel-plate) with gadolinium as the magnetocaloric material (Fig. 2). From the thermo-hydraulic point of view these two structures represent two contrasting examples. The packed-bed structures have relatively good heat-transfer characteristics and large viscous losses, while the parallel-plate structures have relatively poor heat-transfer characteristics and small viscous losses (Šarlah et al., 2012).

Table 1 shows the variables that were analyzed during the geometrical optimization. Different *AMR*s were analyzed at two operating frequencies (0.5 Hz and 3 Hz) and at 40 different mass-flow rates in order to find the optimum mass-flow rate for each analyzed geometry and each operating frequency.

The analyses of the operating parameters and further the geometrical optimization were in the case of the packed-bed *AMR* performed for different lengths of the *AMR* and different sphere diameters. We assumed that the sphere diameter does not affect the porosity of the regenerator and remains constant for all the sphere diameters ($\varepsilon = 0.39$). This is true for relatively small sphere diameters (a large number of particles and a uniform particles size) with respect to the

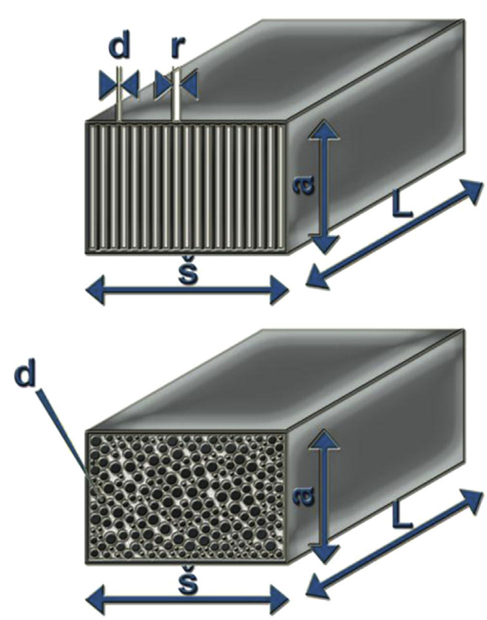

Fig. 2 – Schematic views of the parallel-plate (top) and packed-bed (bottom) *AMR*s.

Table 1 – The analyzed operational and geometrical *AMR* parameters.

Variable/geometry	Spheres	Plates
ν [Hz]	0.5; 3	
$\dot{m}$ [kg s^{-1}]	0.0005–0.3	
L [mm]	20–120	40–200
d [mm]	0.025–2	0.1–1
r [mm]	/	0.01–1
ε [/]	0.39	0.019–0.909
m [kg]	0.0375–0.3	0.022–0.524
A_{ht} [m^2]	0.014–2.278	0.028–0.564

Table 2 – The parameters and properties that remained unchanged during the geometrical analysis.

Magnetic field change	0.0001–1 T
Ratio of the (de)magnetization time and the time of homogenous magnetic field	P_{mag}:P_f = 1:4
Magnetocaloric material	Gd
Outer dimensions of the *AMR*	40 mm × 10 mm × length
Heat-transfer fluid	Water
Temperature span	15 K (T_h = 293 K; T_c = 278 K)
AMR thermodynamic cycle	Brayton-like

regenerator volume (Kaviany, 1995). The mass of the magnetocaloric material in the *AMR*, as a result, does not depend on the sphere diameter. However, the sphere diameter strongly affects the thermo-hydraulic properties of the *AMR*, i.e., the heat-transfer coefficient (Nusselt number) and the friction-factor (Eqs. (3) and (5)).

The same analysis was also performed for different geometries of the parallel-plate *AMR*. We assumed that the plates are oriented parallel to the magnetic field (along the height of the *AMR*, as shown in Fig. 2 (top)). Due to the assumption of a homogenous magnetic field in the magnetocaloric material (neglected demagnetization effect) the orientation of the plates (parallel or perpendicular to the magnetic field) affects only the Graetz number (and consequently the Nusselt number and the heat-transfer coefficient – see Eq. (4)) and not the internal magnetic field. In general, the internal magnetic field and consequently the magnetocaloric effect are strongly affected by the orientation of the plates of the magnetocaloric material in the magnetic field, as recently shown by Nielsen et al. (2012a). However, we analyzed different thicknesses of the plates and different spacings between the plates. Different combinations of plate thicknesses and spacings between the plates provide different porosities for the *AMR* as well as different masses of magnetocaloric material. The porosity, which is defined as the ratio of the empty volume of the *AMR* (occupied by the fluid) with the total volume of the *AMR*, also strongly affects the thermo-hydraulic properties of the *AMR* (heat-transfer coefficient and friction-factor).

Furthermore, we also analyzed the impact of the length of the parallel-plate *AMR* in the case of four different (typical) porosities.

Overall, approximately 7000 different simulations were performed, making this is one of the most comprehensive analyses of the operational and geometrical properties of an *AMR* performed so far. More simulations were recently performed in the studies published by Nielsen et al. (2010) and Bjork and Engelbrecht (2011) where they analyzed the performance of the parallel-plate *AMR* and the influence of the magnetic field on the performance of an *AMR*, respectively. The analyses of the geometrical properties of the *AMR* were recently performed with a steady-state model (via a second-law analysis) by Li et al. (2008) for a packed-bed *AMR* and by Kitanovski et al. (2012) for an *AMR* in the form of a wavy structure.

The parameters and properties that remained unchanged during this analysis are presented in Table 2. Gadolinium was used as the magnetocaloric material, while water was used as the heat-transfer fluid (its properties were considered at 20 °C). We assumed a change of the magnetic field between $1 \cdot 10^{-4}$ T (demagnetization area) and 1 T (magnetization area). We further assumed that the (de)magnetization time is equal to one-fourth of the time of the homogenous magnetic field (also equal to the duration of the fluid flow period) for all analyzed operational frequencies. The impact of this time ratio was studied in detail in Bjork and Engelbrecht (2011). For the purposes of the analysis the predefined temperature span of the *AMR* was set to 15 K (between the hot-side fluid-inlet temperature of 293 K and the cold-side fluid-inlet temperature of 278 K). All the simulations were performed with the regenerative Brayton-like thermodynamic cycle (magnetization–fluid flow–demagnetization–fluid flow in a counter-flow direction). The width ($\check{s}$) and the height (a) of the *AMR* were defined by 40 mm and 10 mm, respectively (see Fig. 2). If we take into account the assumptions of a homogenous distribution of the magnetocaloric material in the *AMR* (neglected flow maldistribution) and neglect the demagnetization effect (homogenous distribution of the magnetic field) these two dimensions do not affect the thermo-hydraulic properties of the 1-D model of the *AMR*. However, the external dimensions of the *AMR*, which are defined by the air gap of the magnet assembly (or vice versa), strongly influence the generated magnetic field and/or the mass of the permanent magnets used, but this analysis was not considered for the purposes of this study (for details see, e.g., Bjork et al., 2008).

Different *AMR* geometries were compared with regards to the specific cooling load (per mass of magnetocaloric material) and the *COP* at the optimum mass-flow rate for each analyzed geometry and operational frequency. The *COP* is defined as the ratio of the cooling load to the input magnetic work and the work needed to pump the fluid (for details see Tušek et al. (2011)).

3.2. *Optimization of a packed-bed* AMR

In order to clearly present the impact of both the geometrical properties of the packed-bed *AMR* (the sphere diameter and the length of the *AMR*) we combined them into the dimensionless length ($L\ d^{-1}$). Fig. 3 shows the maximum specific cooling load and the maximum *COP* as a function of the dimensionless length for different lengths of the *AMR* and two different operating frequencies. For each analyzed *AMR* geometry and operating frequency only the maximum values of the specific cooling load and the maximum values of the *COP*,

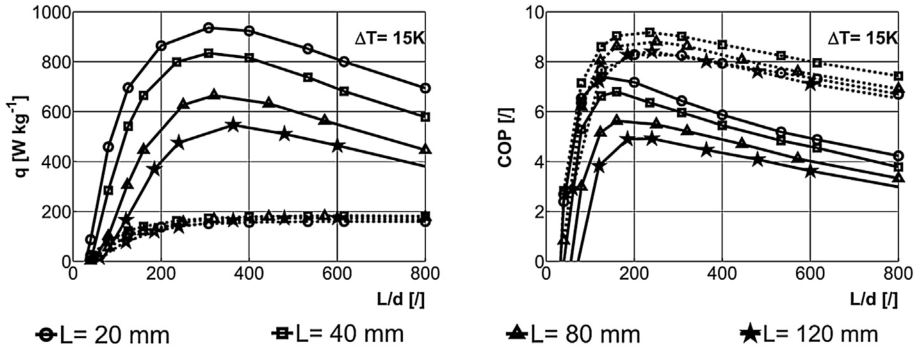

Fig. 3 – The maximum specific cooling load (left) and the maximum *COP* (right) as a function of the dimensionless length for two different operating frequencies (dotted lines for 0.5 Hz and full lines for 3 Hz) and different lengths of the packed-bed *AMR*.

obtained at the optimum mass-flow rate, are shown. The optimum mass-flow rate strongly depends on the geometry of the *AMR*. So, each analyzed *AMR* has a different optimum mass-flow rate. The specific cooling load is included due to it being a more relevant comparison; since different *AMR*s geometries contain different amount of magnetocaloric material.

It is clear that there is an optimum dimensionless length and an optimum sphere diameter for each length of the *AMR* from the cooling load as well as from the *COP* point of view. For the low values of the dimensionless length the heat-transfer losses dominate, since in this case the *AMR* is filled with larger spheres, which leads to a smaller heat-transfer coefficient and a smaller total heat-transfer area. On the other hand, for the higher dimensionless lengths the viscous losses are dominant, which causes decreasing of the specific cooling load and the *COP*. The viscous losses are increasing with the increased length of the *AMR* and a smaller sphere diameter.

We can conclude that the optimum dimensionless length is not significantly affected by the length of the packed-bed *AMR*. In the case of a high operating frequency (3 Hz) the optimum dimensionless length from the cooling-load point of view is approximately 300 and from the *COP* point of view it is between 100 and 200. At a low operating frequency (0.5 Hz) the cooling load is not affected by the dimensionless length above a value of 300 (due to the smaller optimum mass-flow rate), while from the *COP* point of view the optimum dimensionless length is approximately 250.

It is further evident that at a high operating frequency the packed-bed *AMR* shows the best performance for a length of 20 mm (or even less according to the presented trend of dependency), while at a low operating frequency the best performance is for a length of 40 mm (from the specific cooling load and the *COP* point of view). The corresponding optimum sphere diameter (at the optimum lengths) from the cooling-load point of view is approximately 0.07 mm and from the *COP* point of view, approximately 0.17 mm, regardless of the operating frequency.

The difference in the optimum geometry for different frequencies occurs mostly due to the fact that at low frequencies the optimum mass-flow rate and consequently the viscous losses are to some extent smaller than in the case of a high operating frequency. As a result, at a lower operating frequency the optimum *AMR* geometry is different (e.g., smaller optimum sphere diameter, larger optimum dimensionless length) as compared to a higher operating frequency. Furthermore, the viscous losses influence the *COP* to a much

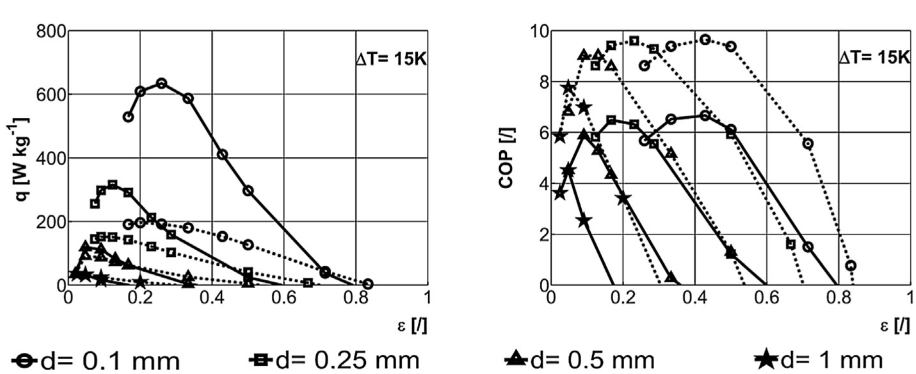

Fig. 4 – The maximum specific cooling load (left) and the maximum COP (right) as a function of the porosity of the parallel-plate *AMR* for two different operating frequencies (dotted lines for 0.5 Hz and full lines for 3 Hz) and different thicknesses of the gadolinium plate.

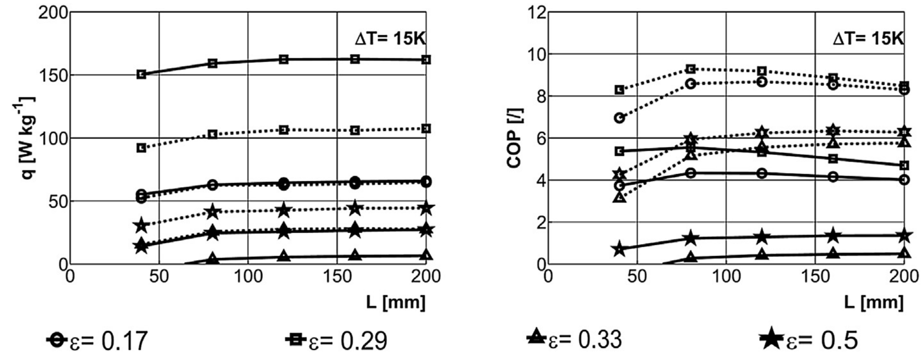

Fig. 5 – The maximum specific cooling load (left) and the maximum *COP* (right) as a function of the length of the parallel-plate *AMR* for two different operating frequencies (dotted lines for 0.5 Hz and full lines for 3 Hz) and four different porosities.

larger extent than the cooling load. This causes a smaller optimum dimensionless length and a smaller optimum sphere diameter from the cooling-load point of view than from the *COP* point of view.

3.3. *Optimization of the parallel-plate* AMR

Fig. 4 shows the maximum specific cooling load and the maximum *COP* (again at the optimum mass-flow rates) as a function of the porosity of the parallel-plate *AMR* for four different thicknesses of the plates. We can see that the parallel-plate *AMR* shows the largest cooling load in the case of a plate that is as thin as possible and that each plate thickness has a different optimum porosity. Such a trend was somehow expected, since thin plates provide a larger total heat-transfer area, while the porosity influences the hydraulic diameter, which is closely related to the heat-transfer coefficient as well as to the friction-factor. At lower porosities the viscous losses dominate, while at higher porosities the heat-transfer losses dominate. It is also evident that increasing the *COP* with a decreased plate thickness is limited to a thickness of 0.25 mm.

We can see that the thinner the magnetocaloric plate is, the larger the optimum porosity will be, while the optimum spacing is in general not dependent on the plate's thickness (only the optimum porosity is). At a high operating frequency the optimum spacing from the *COP* point of view is approximately 0.075 mm, and approximately 0.035 mm from the cooling-load point of view, regardless of the plate's thickness. In the case of a low operating frequency the optimum values of the spacing are slightly smaller, due to the smaller optimum mass-flow rate and the smaller viscous losses, as explained previously for the packed-bed *AMR*, where this impact is even more significant due to larger viscous losses.

If we compare the specific cooling load and the *COP* at both analyzed operating frequencies we can see that higher frequencies, in general, generate a larger specific cooling load and a smaller *COP*. This was expected, since higher operating frequencies are related to larger viscous losses as well as larger heat-transfer losses. Such a trend was also noted in the case of the packed-bed *AMR*. However, for the parallel-plate *AMR* in the case of the thicker plates (more than 0.5 mm) a larger specific cooling load is generated with a lower frequency, since a longer time is needed (lower frequency) for the fluid to absorb all the energy from the magnetocaloric plate.

Fig. 5 shows the maximum specific cooling load and the maximum *COP* (at the optimum mass-flow rates) as a function of the length of the parallel-plate *AMR* for four typical porosities (16.7% (d = 0.5 mm; r = 0.1 mm), 28.6% (d = 0.25 mm; r = 0.1 mm), 33.3% (d = 0.5 mm; r = 0.25 mm) and 50% (d = 0.25 mm; r = 0.25 mm)). It is evident that due to the relatively small viscous losses the length does not have a significant impact on the specific cooling load. The exceptions are the *AMR*s with a shorter length (less than 80 mm), since such *AMR*s cannot exceed the predefined temperature span (15 K). Furthermore, from the *COP* point of view we can conclude that in the case of smaller porosities the *COP* is decreasing with the length of the *AMR* due to the impact of the viscous losses (for the lengths longer than 80 mm, at which the *AMR* exceeds the predefined temperature span). In the case of higher porosities the length does not influence the *AMR*'s efficiency.

Table 3 – The optimized geometrical properties of the packed-bed and parallel-plate *AMR*.

	Low frequency (0.5 Hz)		High frequency (3 Hz)	
	With regards to the q	With regards to the COP	With regards to the q	With regards to the COP
Packed-bed *AMR*	$L = 40$ mm $d = 0.07$ mm	$L = 40$ mm $d = 0.17$ mm	$L \leq 20$ mm $d = 0.07$ mm	$L \leq 20$ mm $d = 0.17$ mm
Parallel-plate *AMR*	$d \leq 0.1$ mm $s = 0.035$ mm	$d \leq 0.25$ mm $s = 0.075$ mm	$d \leq 0.1$ mm $s = 0.035$ mm	$d \leq 0.25$ mm $s = 0.075$ mm

4. Conclusions

A comprehensive numerical analysis was performed in order to analyze the geometrical properties of packed-bed and parallel-plate *AMR*s. The main results of the work are presented in Table 3, where the optimum values of all evaluated geometrical properties (except the length of the parallel-plate *AMR*) are shown with regards to the specific cooling load and the *COP*.

We can conclude that the packed-bed *AMR* has an optimum sphere diameter for each length of the *AMR*. The shorter the *AMR* is, the smaller the sphere diameter needs to be to ensure the best cooling characteristics (cooling load and *COP*). For each operating frequency there is an optimum dimensionless length ($L\ d^{-1}$), which is relatively independent of the *AMR*'s length and different from the specific cooling load and from the *COP* point of view. In order to generate larger cooling loads and *COP* values and to minimize the use of the magnetocaloric material, we should consider a wider and shorter (20 mm or less from the cooling-load point of view and 40 mm from the *COP* point of view) packed-bed *AMR* filled with the optimum corresponding sphere diameter (approximately 0.07 mm from the cooling-load point of view and approximately 0.17 mm from the *COP* point of view).

The parallel-plate *AMR* should be constructed with plates that are as thin as possible and the optimum corresponding porosity (spacing). The optimum spacing for the parallel-plate *AMR* is between 0.035 mm (from the cooling-load point of view) and 0.075 mm (from the *COP* point of view), regardless of the plate thickness and in general also regardless of the operational frequency. Furthermore, due to the relatively small viscous losses of the parallel-plate structures, the length of the parallel-plate *AMR* does not significantly affect the specific cooling load (for lengths longer than 80 mm), while the *COP* is slightly reduced when a structure with small porosities is considered. Therefore, in order to generate higher cooling loads, *COP*s and temperature spans at higher porosities, it is recommended to use the longer parallel-plate *AMR* (between 80 and 160 mm). In the case of lower porosities the optimum length is approximately 80 mm.

Based on the above-presented results we can further conclude that the packed-bed *AMR* is able generate a higher specific cooling load (especially at higher operating frequencies), which is due to the better heat-transfer characteristics (larger total heat-transfer area and smaller hydraulic diameter). It is to be expected that the parallel-plate *AMR* with a plate thickness below 0.1 mm would generate specific cooling loads comparable to the packed-bed *AMR*s. On the other hand, the parallel-plate *AMR* ensures slightly better *COP* values, mostly due to the smaller viscous losses. Shorter packed-bed *AMR*s can operate with similar efficiency as the parallel-plate *AMR*.

It is to be expected that, e.g., different magnetocaloric materials, different magnetic field changes, different heat-transfer fluids or different temperatures spans of the *AMR* would result in potentially different optimum geometrical properties, since different mass-flow rates would be required for optimum operation. For example, if the analyzed *AMR*s are subjected to a larger magnetic field change, a larger optimum mass-flow rate would be required due to larger adiabatic temperature change of magnetocaloric material. This would further cause larger viscous losses, which would result in a larger optimum sphere diameter for the packed-bed *AMR* and a larger optimum spacing between the plates for the parallel-plate *AMR*. The optimum *AMR* geometry is therefore also related to other operating and magnetic properties, so it should be properly considered. However, with the nowadays known magnetocaloric materials and the magnetic fields generated by the permanent magnets the optimum geometries are expected to be relatively close to the values presented in Table 3.

It should be pointed out that the development of a parallel-plate *AMR* with a plate thickness of 0.1 mm or less and a spacing of approximately 0.05 mm also from other magnetocaloric materials is one of the major challenges for magnetic refrigeration at room temperature in the near future. This is very important, especially for operation at higher frequencies, which enables higher cooling loads. Furthermore, it is necessary that the *AMR*s are constructed with none or minor maldistribution, since this can drastically affect the *AMR*'s performance.

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